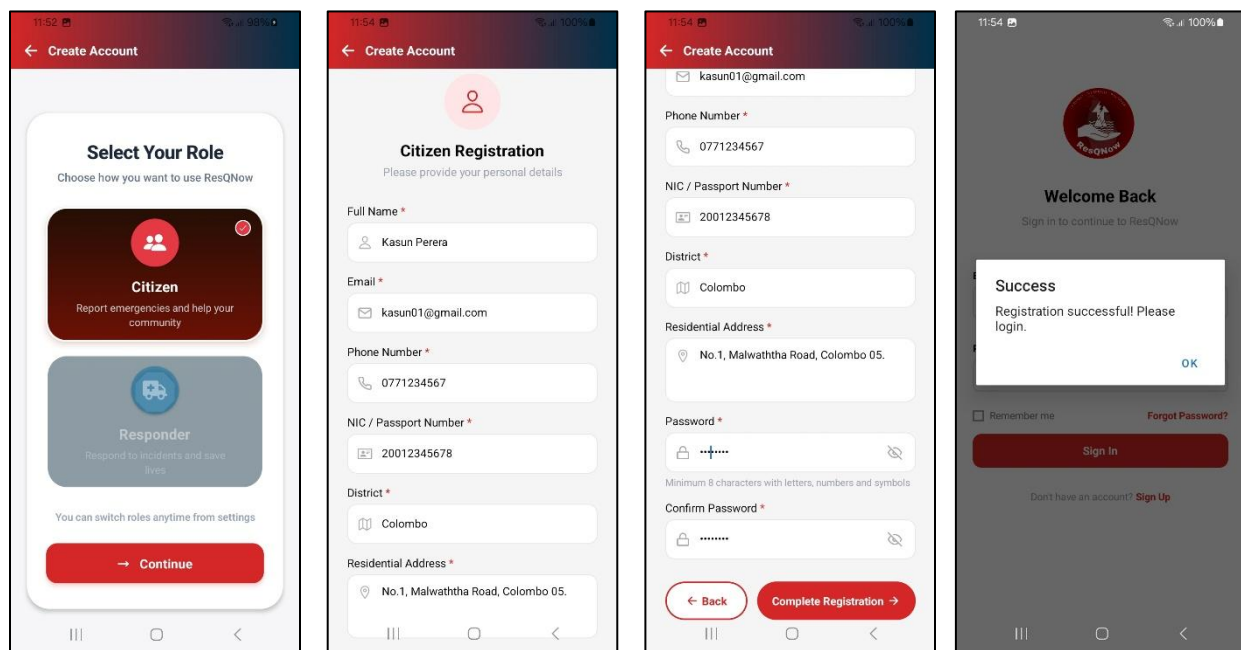


Authentication Module Test Cases

- AUTH-001 Citizen Registration
- AUTH-002 Duplicate Email
- AUTH-003 Citizen Login
- AUTH-004 Wrong Password
- AUTH-005 Forgot Password (Citizen)
- AUTH-006 Forgot Password (Responder)
- AUTH-007 Forgot Password (Admin)
- AUTH-008 Verify OTP
- AUTH-009 Reset Password
- AUTH-010 Expired OTP
- AUTH-011 Invalid OTP
- AUTH-012 Responder Registration
- AUTH-013 Responder Pending Verification
- AUTH-014 Verified Responder Login

Field	Details / Output
Test ID	AUTH-001
Module	Authentication
Feature	Citizen Registration
Actual Result	User registered successfully. Account created in MongoDB with hashed password. Backend returned HTTP 201 Created in 120 ms. UI navigated to Login.
Status	PASS



<pre> _id: ObjectId('6a7428b28a37c6808ed5dfa5') name: "Kasun Perera" email: "kasun01@gmail.com" password: "\$2b\$10\$QnReiDYsxW6psqCn14if00LACBkfSggzPnumLZ5XY61iKs0ZJZAK" role: "Citizen" district: "Colombo" location: Object organization: "-" contact_number: "0771234567" profileImage: null notificationSettings: Object privacySettings: Object twoFactorEnabled: false shareLocation: true shareReports: true profileVisibility: "public" isVerified: true status: "Active" documents: Object registered_date: 2026-08-06T06:24:50.749+00:00 createdAt: 2026-08-06T06:24:50.754+00:00 updatedAt: 2026-08-06T06:24:50.754+00:00 __v: 0 </pre>
--

Field	Details / Output
Test ID	AUTH-002
Module	Authentication
Feature	Duplicate Email Handling
Actual Result	Registration blocked. Backend returned HTTP 400 Bad Request (or 409 Conflict) with "User with this email already exists". UI displayed proper error alert. No duplicate document added to MongoDB.
Status	PASS

Create Account

Citizen Registration
Please provide your personal details

Full Name *
Kasun Perera

Email *
kasun01@gmail.com

Phone Number *
0779998887

NIC / Passport Number *
20009998887

District *
Gampaha

Residential Address *
No. 12, Main Road, Ja Ela.

Create Account

kasun01@gmail.com

Phone Number *
0779998887

NIC / Passport Number *
20009998887

District *
Gampaha

Residential Address *
No. 12, Main Road, Ja Ela.

Password *
Test@456

Minimum 8 characters with letters, numbers and symbols

Confirm Password *
Test@456

Back Complete Registration

Create Account

kasun01@gmail.com

Phone Number *
0779998887

NIC / Passport Number *
20009998887

District *
Gampaha

Residential Address *
No. 12, Main Road, Ja Ela.

Password *
Test@456

Minimum 8 characters with letters, numbers and symbols

Confirm Password *
Test@456

Back Complete Registration

Error
User already exists
OK

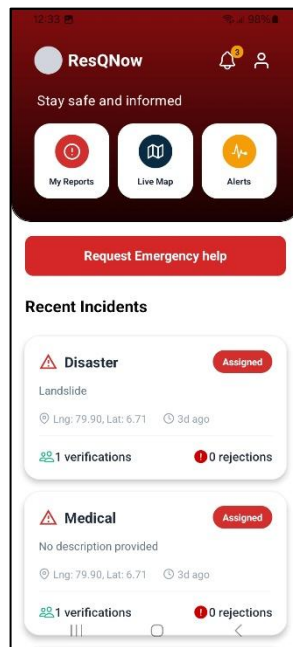
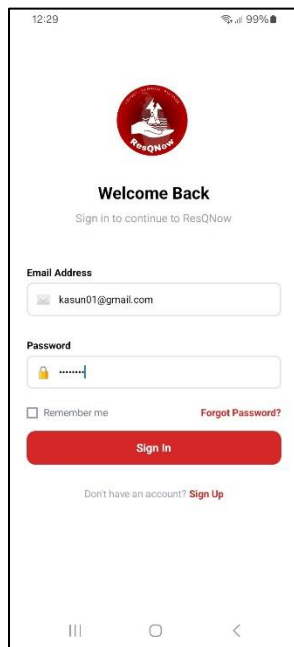
```

PS G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend> node server.js
[dotenv@17.3.1] injecting env (8) from .env -- tip: ⚙ enable debug logging with { debug: true }
MongoDB Connected ✅
🚀 Unified real-time server deployed publicly on port 5000

-----
📄 Incoming API Request: [POST] /api/auth/register
🕒 Time: 12:22:51 PM
-----
POST /api/auth/register 400 33 - 158.847 ms

```

Field	Output Details
Test ID	AUTH-003
Module	Authentication
Feature	Citizen Login
Actual Result	User logged in successfully. Backend returned HTTP 200 OK with valid JWT token in ~105 ms. UI navigated to Citizen Dashboard.
Status	PASS



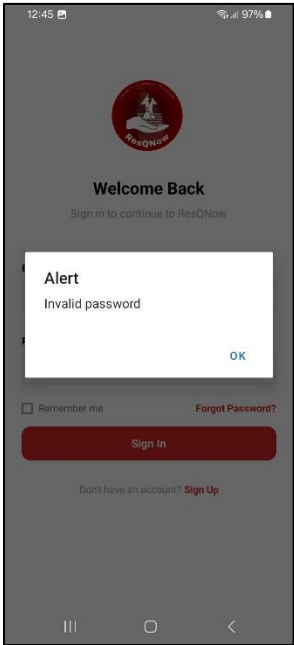
```

-----
Incoming API Request: [GET] /api/auth/profile
Time: 12:33:22 PM
-----
GET /api/auth/profile 200 907 - 103.416 ms

-----
Incoming API Request: [GET] /api/incidents?page=1&limit=10
Time: 12:33:23 PM
-----
Socket connection established with device token: d03ncHww3_2rhHDZAAAB
Secure communications room mapped for User ID: 6a7428b28a37c6808ed5dfa5
GET /api/incidents?page=1&limit=10 304 - - 102.847 ms

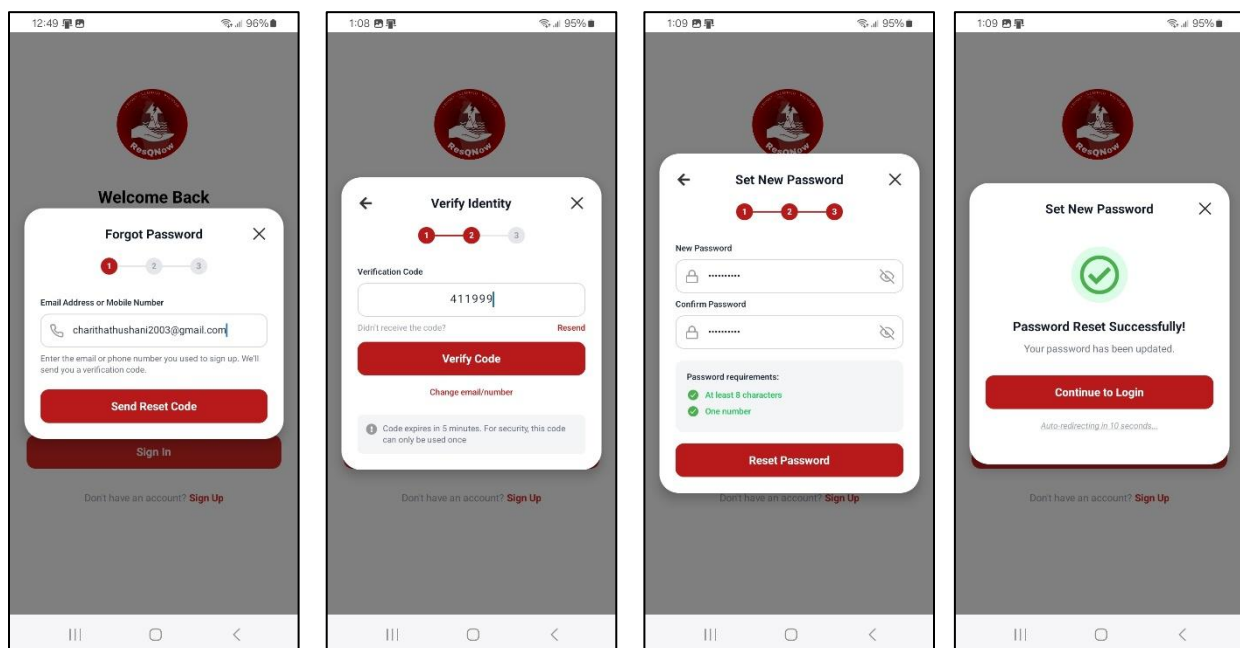
```

Field	Output Details
Test ID	AUTH-004
Module	Authentication
Feature	Invalid Password Handling
Actual Result	Login rejected. Backend returned HTTP 400 / 401 with error message "Invalid email or password" in ~175 ms. App displayed error toast and blocked dashboard access.
Status	PASS



```
-----
Incoming API Request: [POST] /api/auth/login
Time: 12:48:09 PM
-----
POST /api/auth/login 400 30 - 176.502 ms
█
```

Field	Output Details
Test ID	AUTH-005, AUTH-008, AUTH-009
Module	Authentication
Feature	Forgot Password - Citizen (OTP Request, Verify OTP, Reset Password)
Actual Result	Password reset request processed successfully. OTP generated and sent to target email. Backend returned HTTP 200 OK in ~240 ms. Password Reset successfully.
Status	PASS

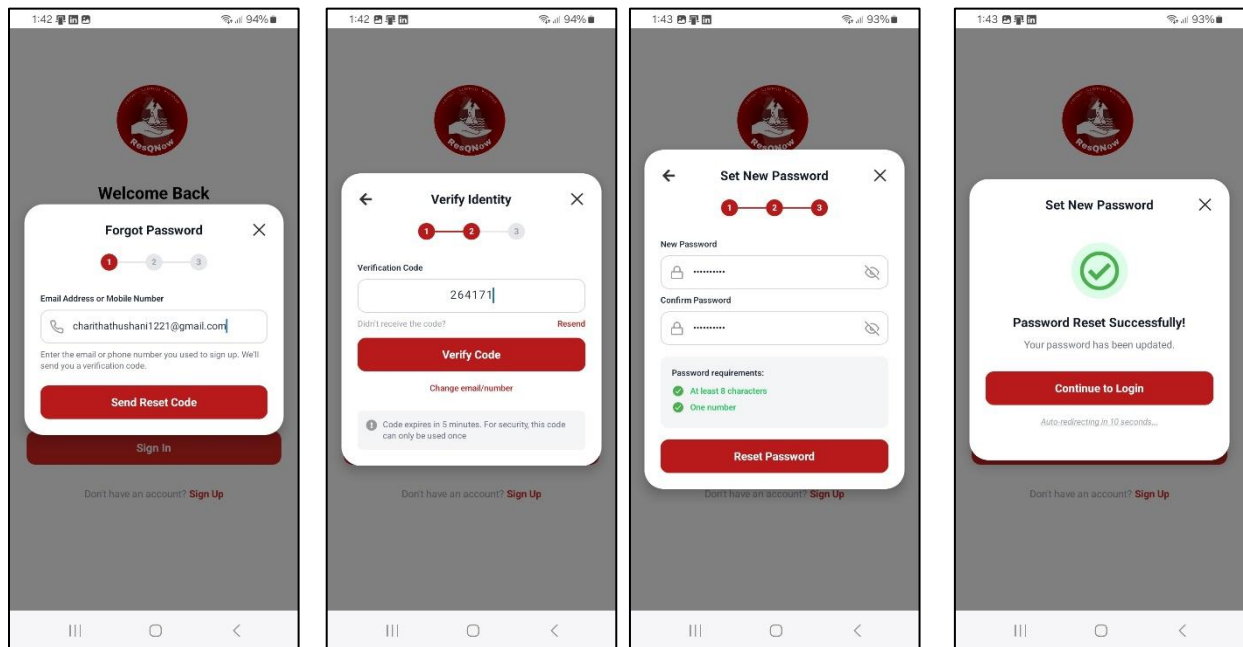


```

-----
Incoming API Request: [POST] /api/auth/forgot-password
Time: 1:08:16 PM
-----
[OTP SERVICE]: Generated OTP 411999 for target user: charithathushani2003@gmail.com
EMAIL SENT
To: charithathushani2003@gmail.com
Subject: Password Reset OTP
Message: Your OTP is: 411999
POST /api/auth/forgot-password 200 35 - 219.893 ms
Socket connection established with device token: SWx3sDVmWwWw250hAAAB
Secure communications room mapped for User ID: 6a7428b28a37c6808ed5dfa5
-----
Incoming API Request: [POST] /api/auth/verify-otp
Time: 1:08:45 PM
-----
POST /api/auth/verify-otp 200 26 - 85.247 ms
-----
Incoming API Request: [POST] /api/auth/reset-password
Time: 1:09:16 PM
-----
POST /api/auth/reset-password 200 39 - 277.074 ms
Device connection dropped: transport close

```

Field	Details / Output
Test ID	AUTH-006, AUTH-008, AUTH-009
Module	Authentication
Feature	Forgot Password - Responder (OTP Request, Verify OTP, Reset Password)
Actual Result	OTP validated successfully against stored database hash/value. Backend returned HTTP 200 OK in ~200 ms. Password Reset successfully.
Status	PASS



 Incoming API Request: [POST] /api/auth/forgot-password

Time: 1:42:32 PM

[OTP SERVICE]: Generated OTP 264171 for target user: charithathushani1221@gmail.com

EMAIL SENT

To: charithathushani1221@gmail.com

Subject: Password Reset OTP

Message: Your OTP is: 264171

POST /api/auth/forgot-password 200 35 - 1078.226 ms

Incoming API Request: [POST] /api/auth/verify-otp

Time: 1:42:45 PM

POST /api/auth/verify-otp 200 26 - 82.319 ms

Incoming API Request: [POST] /api/auth/reset-password

Time: 1:43:23 PM

POST /api/auth/reset-password 200 39 - 207.582 ms

Field	Details / Output
Test ID	AUTH-007, AUTH-008, AUTH-009
Module	Authentication
Feature	Forgot Password - Admin (OTP Request, Verify OTP, Reset Password)
Actual Result	OTP validated successfully against stored database hash/value. Backend returned HTTP 200 OK in ~250 ms. Password Reset successfully.
Status	PASS

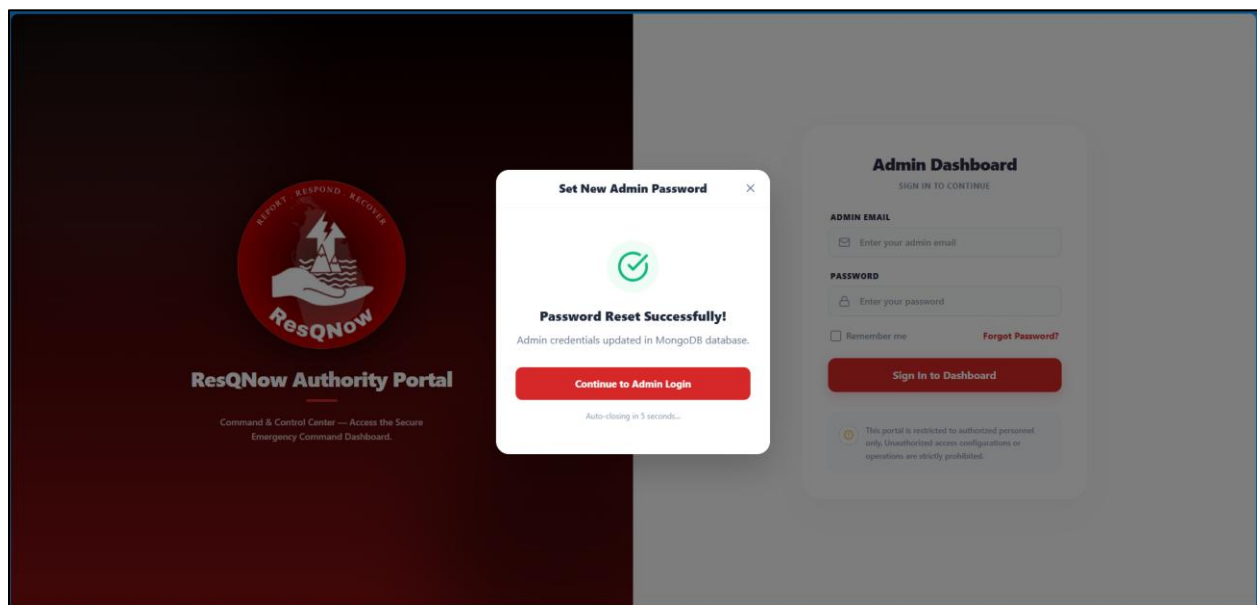
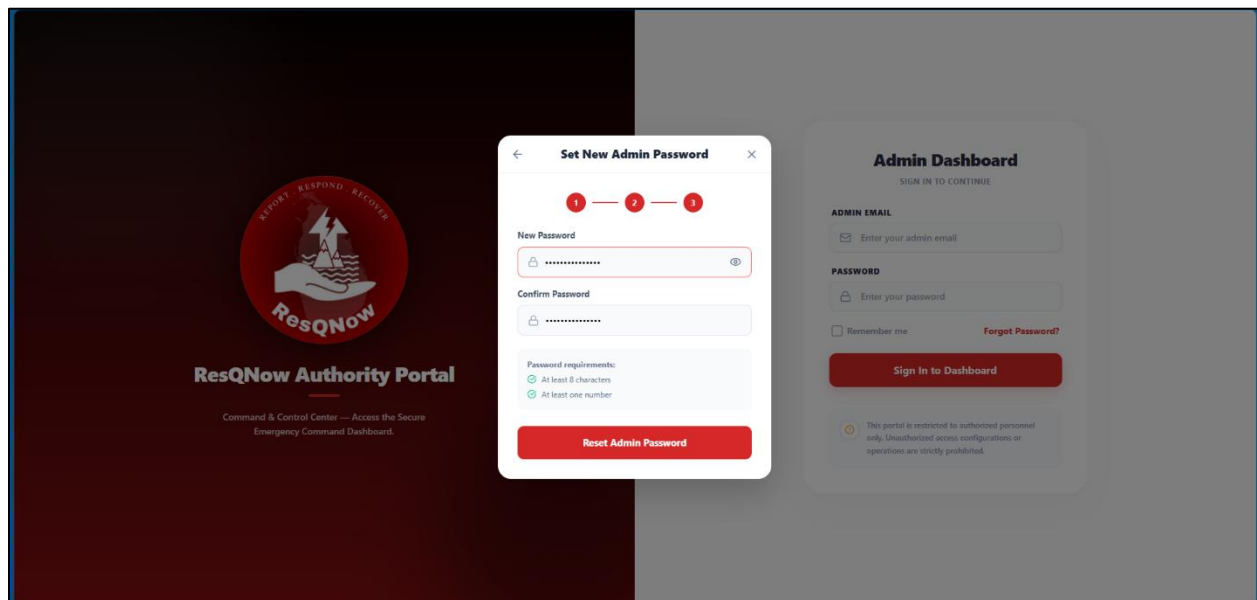
The screenshot displays the ResQNow Authority Portal interface. On the left, the portal logo features a red circle with a white hand holding a green leaf, surrounded by the text 'REPORT RESPOND RECOVER' and 'ResQNow'. Below the logo, the text reads 'ResQNow Authority Portal' and 'Command & Control Center — Access the Secure Emergency Command Dashboard.'.

In the center, a modal titled 'Admin Security Reset' is open. It has a progress bar with three steps: 1 (selected), 2, and 3. The first step, 'Admin Email Address', shows the email 'superadmin@test.com'. The second step, 'Master System Key', shows a masked key '*****'. Below the key, a note states: 'Provide your registered Admin email along with the System Master Key to authorize reset code dispatch.' A red button labeled 'Authorize & Send OTP' is at the bottom of the modal.

On the right, the 'Admin Dashboard' is visible, titled 'Admin Dashboard' with the subtitle 'SIGN IN TO CONTINUE'. It contains input fields for 'ADMIN EMAIL' (with a placeholder 'Enter your admin email') and 'PASSWORD' (with a placeholder 'Enter your password'). There is a 'Remember me' checkbox and a 'Forgot Password?' link. A red button labeled 'Sign In to Dashboard' is below the password field. A warning message at the bottom states: 'This portal is restricted to authorized personnel only. Unauthorized access configurations or operations are strictly prohibited.'

This screenshot shows the 'Verify Admin Identity' modal on the ResQNow Authority Portal. The modal has a progress bar with three steps: 1, 2 (selected), and 3. The second step, 'Verification Code', displays the code '8 7 0 1 3 4'. Below the code, there is a 'Didn't receive code?' link and a 'Resend in 17s' timer. A red button labeled 'Verify Security Code' is at the bottom of the modal. Below the button, there is a link 'Change admin email' and a note: 'Code expires in 3 minutes for elevated security. Code can only be used once.'

The background shows the same 'Admin Dashboard' as the previous screenshot, with the 'ADMIN EMAIL' and 'PASSWORD' fields, 'Remember me' checkbox, 'Forgot Password?' link, and 'Sign In to Dashboard' button. The warning message at the bottom remains the same.



🔑 [ADMIN SECURITY]: Master Key Verified!
📧 [OTP SERVICE]: Admin Reset OTP 870134 issued for superadmin@test.com
EMAIL SENT
To: superadmin@test.com
Subject: CRITICAL: Admin Password Reset OTP
Message: High-Security Action Requested. Your Secure Admin OTP is: 870134. This code expires in 3 minutes.
POST /api/auth/admin-forgot-password 200 95 - 337.520 ms

📡 Incoming API Request: [POST] /api/auth/verify-otp
🕒 Time: 2:11:41 PM

POST /api/auth/verify-otp 200 26 - 95.586 ms

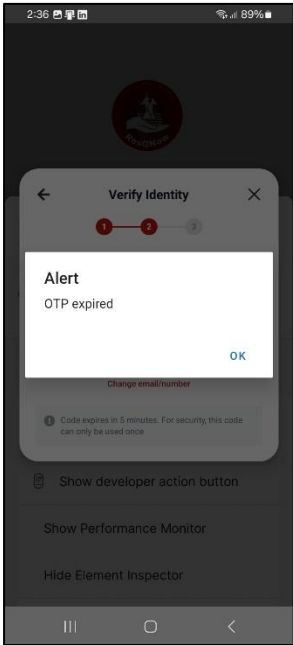
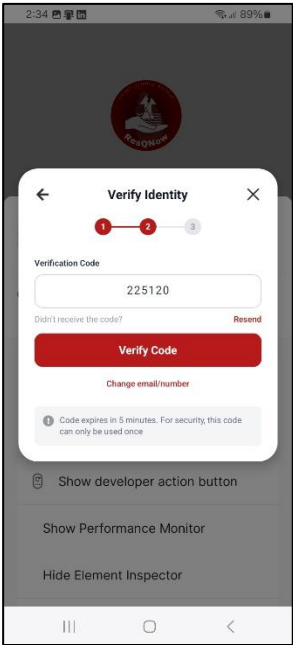
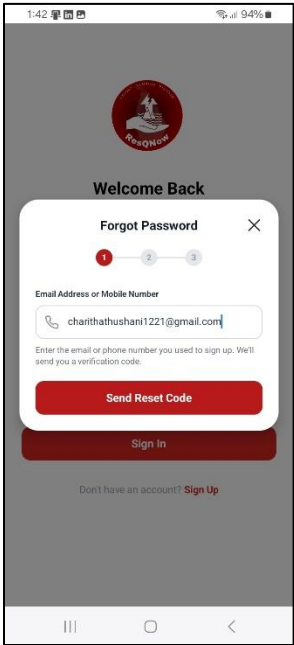
📡 Incoming API Request: [POST] /api/auth/reset-password
🕒 Time: 2:13:09 PM

POST /api/auth/reset-password 200 39 - 539.014 ms

📡 Incoming API Request: [POST] /api/auth/login
🕒 Time: 2:14:58 PM

POST /api/auth/login 200 247 - 427.042 ms

Field	Details / Output
Test ID	AUTH-010
Module	Authentication
Feature	Forgot Password - Expired OTP Handling
Actual Result	Backend identified expired timestamp (otpExpire < Date.now()) and returned HTTP 400 Bad Request ("OTP expired") in ~210 ms. UI displayed error alert requesting a new code dispatch.
Status	PASS



```
-----
Incoming API Request: [POST] /api/auth/forgot-password
Time: 2:29:27 PM
-----

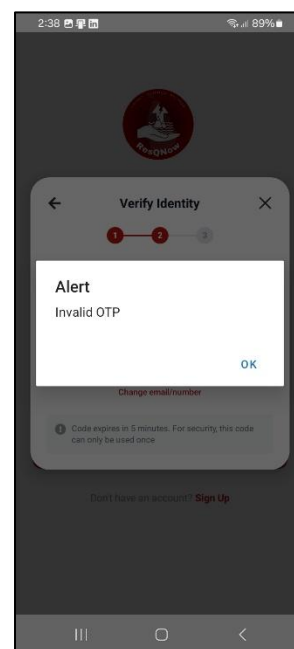
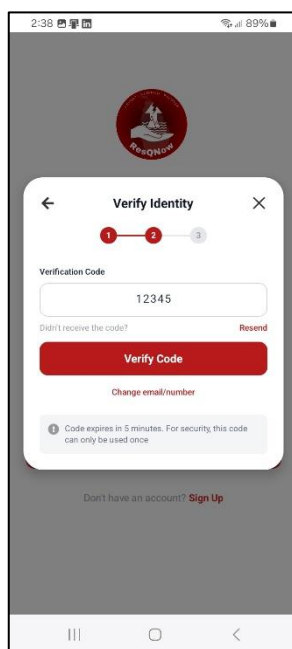
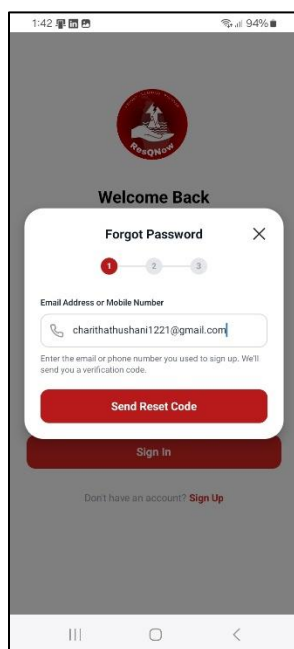
[OTP SERVICE]: Generated OTP 225120 for target user: charithathushani1221@gmail.com
EMAIL SENT
To: charithathushani1221@gmail.com
Subject: Password Reset OTP
Message: Your OTP is: 225120
POST /api/auth/forgot-password 200 35 - 209.222 ms

-----

Incoming API Request: [POST] /api/auth/verify-otp
Time: 2:36:06 PM
-----

POST /api/auth/verify-otp 400 25 - 81.263 ms
█
```

Field	Details / Output
Test ID	AUTH-011
Module	Authentication
Feature	Forgot Password - Invalid OTP Validation
Actual Result	Backend compared submitted code against stored database OTP, rejected the mismatch, and returned HTTP 400 Bad Request ("Invalid OTP") in ~98 ms. UI displayed error banner without advancing step.
Status	PASS



```

-----
Incoming API Request: [POST] /api/auth/forgot-password
Time: 2:38:33 PM
-----

[OTP SERVICE]: Generated OTP 653827 for target user: charithathushani1221@gmail.com
EMAIL SENT
To: charithathushani1221@gmail.com
Subject: Password Reset OTP
Message: Your OTP is: 653827
POST /api/auth/forgot-password 200 35 - 197.673 ms

-----

Incoming API Request: [POST] /api/auth/verify-otp
Time: 2:38:48 PM
-----

POST /api/auth/verify-otp 400 25 - 97.359 ms

```

Field	Details / Output
Test ID	AUTH-012
Module	Authentication
Feature	Responder Registration & Document Upload
Actual Result	Registration payload and verification documents uploaded to Cloudinary. Account created with status "Pending" and isVerified: false. Admin notification email dispatched. Backend returned HTTP 201 Created in ~5000 ms.
Status	PASS

2:45

Create Account

Select Your Role

Choose how you want to use ResQNow

Citizen

Report emergencies and help your community

Responder

Respond to incidents and save lives

You can switch roles anytime from settings

Continue

2:45

Create Account

Step 1 of 6

Responder Registration

Please provide your personal details

Full Name *

R. Saman Fernando

Email *

fernando@gmail.com

Phone Number *

07766633298

NIC / Passport Number *

200156798754

BackContinue

2:47

Create Account

Step 2 of 6

Organization Details

Tell us about your organization and role

Organization Type *

Fire Department

Organization Name *

Colombo East Fire Center

Phone Number *

0776699852

Official Position / Role *

Chief Officer

Official ID / Employee Number

Emp-201

BackContinue

2:48

Create Account

Step 4 of 6

Credentials & Skills

Share your certifications and expertise

Relevant Certifications *

Select all that apply

First Aid

CPR

Search & Rescue

EMT

Lifeguard

Disaster Response

Years of Experience *

3

Training Completed

LifeGuard Service Training 2024

BackContinue

2:49 87% 2:49 87% 2:49 87%

← Create Account

Step 5 of 6 Location



Location & Availability
Define your working areas and schedule

District *
Colombo

Divisional Secretariat *
Colombo East

Typical Response Area
Borella, Jayawardhanapura Kotte

Availability *
Full Time Part Time On Call
Weekends Only

Work Schedule *
☒ Available on Weekdays
☒ Available on Weekends

Vehicle Ownership (Optional)

2:49 87% 2:49 87% 2:49 87%

← Create Account

Borella, Jayawardhanapura Kotte

Availability *
Full Time Part Time On Call
Weekends Only

Work Schedule *
☒ Available on Weekdays
☒ Available on Weekends

Vehicle Ownership (Optional)
4x4 Boat Bike Ambulance
Truck

Equipment Carried (Optional)
First Aid Kit Rope Rain Gear
Life Jackets Flashlight Radio

Languages Spoken *
English Sinhala Tamil

← Back Continue →

Already have an account? Sign in

2:50 87% 2:50 87% 2:50 87%

← Create Account

Step 6 of 6 Verification



Document Verification
Upload required documents for verification

Important
All documents will be reviewed by our verification team. Please ensure they are clear, readable, and valid.

Password *
saman@1221

Confirm Password *
saman@1221

Official ID / Employee Badge *
Upload a clear photo of your official ID or employee badge
Screenshot_20260806_143609_Expo Go.jpg

Authorization Letter / Appointment Letter *
Letter from your organization authorizing you as a responder
Screenshot_20260806_130919_Expo Go.jpg

Certification Cards
Upload any relevant certification cards (CPR, EMT, First Aid, etc.)
Screenshot_20260806_130506_Expo Go.jpg

← Back Submit for Verification

Already have an account? Sign in

2:50 87% 2:50 87% 2:50 87%

← Create Account

Upload required documents for verification

Important
All documents will be reviewed by our verification team. Please ensure they are clear, readable, and valid.

Password *
saman@1221

Confirm Password *
saman@1221

Official ID / Employee Badge *
Upload a clear photo of your official ID or employee badge
Screenshot_20260806_143609_Expo Go.jpg

Authorization Letter / Appointment Letter *
Letter from your organization authorizing you as a responder
Screenshot_20260806_130919_Expo Go.jpg

Certification Cards
Upload any relevant certification cards (CPR, EMT, First Aid, etc.)
Screenshot_20260806_130506_Expo Go.jpg

← Back Submit for Verification

Already have an account? Sign in

2:51 87% 2:51 87% 2:51 87%



Account Under Review
Thank you for registering as an emergency responder.
Your account is currently under verification.

Application Received
Your registration has been submitted successfully

Under Verification
Our team is reviewing your credentials

Expected Response Time
24-48 hours

You will receive an email at once your account is verified.

Continue

```
_id: ObjectId('6a745227351c896d7ddb6515')
name: "R. Saman Fernando "
email: "fernando@gmail.com"
password: "$2b$10$AUSADbpSsX8Brja10TFYQe007KHtB7K7RVwtMP6Ew1f5rDJaiCy8i"
role: "Responder"
district: "Colombo"
location: Object
organization: "Colombo East Fire Center"
contact_number: "07766633298"
profileImage: null
notificationSettings: Object
privacySettings: Object
isVerified: false
status: "Pending"
documents: Object
registered_date: 2026-08-06T09:21:43.172+00:00
createdAt: 2026-08-06T09:21:43.173+00:00
updatedAt: 2026-08-06T09:21:43.173+00:00
__v: 0
```

Field	Details / Output
Test ID	AUTH-013, AUTH-014
Module	Authentication
Feature	Responder Pending Verification Access Block, Verified Responder Login
Actual Result	Login attempt by unapproved responder blocked by status guard check (status === "Pending"). Backend returned HTTP 403 Forbidden ("Your responder account is still under review") in ~110 ms. Admin approved account (status: "Approved"). Credentials validated via bcrypt, JWT generated, and backend returned HTTP 200 OK in ~160 ms. Client granted access to responder dashboard.
Status	PASS

ResQNow Admin

Dashboard

Incident Management

Verification Center

Responder Management

Danger Zone Management

Analytics & Reports

User Management

System Settings

Alerts Center

Logout

User Management

System Users

Manage system users, roles, and access control

Search by name, email, or ID...

All Roles

Export CSV

☐	USER INFO	ROLE	STATUS	JOINED	ACTIONS
☐	RSF R. Soman Fernando (6a745227351c896d7ddb6515) fernando@gmail.com	Responder	Pending	Aug 06, 2026	Pending Fleet Action
☐	KP Kasun Perera (6a7428b28a37c6808ed5dfa9) kasun01@gmail.com	Citizen	Active	Aug 06, 2026	
☐	TU Test User (6a7223647c65eca49332c4f5) test1785865059199@example.com	Citizen	Active	Aug 04, 2026	
☐	TU Test User (6a721ef90a0f6e0ba058c32a) test1785863926368@example.com	Citizen	Active	Aug 04, 2026	
☐	TU Test User (6a377baca43bb2a8a681d990) test1782053803785@example.com	Citizen	Active	Jun 21, 2026	

Showing 5 entries

Previous1Next

ResQNow Admin

Dashboard

Incident Management

Verification Center

Responder Management

Danger Zone Management

Analytics & Reports

User Management

System Settings

Alerts Center

Logout

Responder Management

JAJohn Anderson
HEAD DISPATCHER

Fleet Directory

Verify submitted credentials and evaluate deployment permissions

+ Add New Responder

Active Fleet (28)

Pending Approvals

Registration Requests Queue

Search incoming requests...

All Departments

REQUEST ID	APPLICANT	CONTACT DETAILS	DISTRICT	AFFILIATED AGENCY	SUBMITTED	VERIFICATION ACTIONS
6a745227351c896d7db6515	R. Saman Fernando View Onboarding Credentials	fernando@gmail.com 0776633298	Colombo	Colombo East Fire Center	11 mins ago	ApproveReject
6a2563f23c879223b6ab5586	Test User View Onboarding Credentials	test1780835312808@example.com 0771234567	Kalutara	Medical	59 days ago	ApproveReject
6a128afcf7b28ec81a2f8f1	DB Silva View Onboarding Credentials	responder1est@gmail.com 0771234525	Colombo	Colombo Police	74 days ago	ApproveReject
6a115138f6499364dc268bf5	D Silva View Onboarding Credentials	responder@gmail.com 0715684256	Colombo	ABD.pvt.ltd	75 days ago	ApproveReject
6a88596a2681bfca8b759254	kavindu View Onboarding Credentials	wkanjana1590@gmail.com 0716053892	galle	police	81 days ago	ApproveReject

ResQNow Admin

Dashboard

Incident Management

Verification Center

Responder Management

Danger Zone Management

Analytics & Reports

User Management

System Settings

Alerts Center

Logout

Responder Management

JAJohn Anderson
HEAD DISPATCHER

Fleet Directory

Verify submitted credentials and evaluate deployment permissions

+ Add New Responder

Active Fleet (28)

Pending Approvals

Registration Requests Queue

Search incoming requests...

All Departments

REQUEST ID	APPLICANT	CONTACT DETAILS	DISTRICT	AFFILIATED AGENCY	SUBMITTED	VERIFICATION ACTIONS
6a745227351c896d7db6515	R. Saman Fernando View Onboarding Credentials	fernando@gmail.com 0776633298	Colombo	Colombo East Fire Center	11 mins ago	ApproveReject
6a2563f23c879223b6ab5586	Test User View Onboarding Credentials	test1780835312808@example.com 0771234567	Kalutara	Medical	59 days ago	ApproveReject
6a128afcf7b28ec81a2f8f1	DB Silva View Onboarding Credentials	responder1est@gmail.com 0771234525	Colombo	Colombo Police	74 days ago	ApproveReject
6a115138f6499364dc268bf5	D Silva View Onboarding Credentials	responder@gmail.com 0715684256	Colombo	ABD.pvt.ltd	75 days ago	ApproveReject
6a88596a2681bfca8b759254	kavindu View Onboarding Credentials	wkanjana1590@gmail.com 0716053892	galle	police	81 days ago	ApproveReject

Application Approved!

Verification email link has been dispatched to the responder.

Closing in 1s...

3:0786%

ResQNow

Welcome Back

Sign in to continue to ResQNow

Email Address

fernando@gmail.com

Password

.....

☐ Remember me

Forgot Password?

Sign In

Don't have an account? Sign Up